MIL-G-1177
22 July 1949

MILITARY SPECIFICATION

GUN, MACHINE, BROWNING, CALIBER .30, M1917A1

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 52-5-1B
4 June 1943

This specification consists of this cover sheet and Specification 52-5-1B, dated 4 June 1943, together with Amendment 2, dated 28 July 1944, attached hereto, without modification.

Copies of this specification may be obtained upon application to the Office, Chief of Ordnance, Washington 25, D. C.

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U. S. ARMY
SPECIFICATION

No. 52-5-1B

JUNE 4, 1943

SUPERSEDING
No. 52–5–1A
August 5, 1939

GUN, MACHINE, BROWNING, CALIBER .30, M1917A1

A. APPLICABLE SPECIFICATIONS.

A–1. Ordinance Department drawing 51–10–41, entitled "List of Drawings and Specifications, Gun, Machine, Caliber .30 Browning, M1917A1", lists the drawings and specifications necessary to complete the manufacture of the machine gun.

A–2. All drawings and specifications listed on the drawings mentioned in paragraph B–1, of the revision date indicated thereon, form a part of this specification.

B. TYPES.

B–1. This specification covers one type of machine gun, as follows:

Gun, Machine, Browning, Caliber .30 M1917A1, Drawing No. 51–10–41

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—Material shall be in accordance with applicable drawings and specifications.

C–2. *Workmanship.*—Finished machine guns shall be free from defects that may affect appearance, serviceability, operation and functioning.

D. GENERAL REQUIREMENTS.

D–1. See detail requirements.

E. DETAIL REQUIREMENTS.

E–1. *Protective Finish.*—

E–1a. Protective finishes shall be as prescribed on the drawings and shall meet the requirements of U. S. Army Specification No. 57–0–2, "Finishes, Protective, on Iron and Steel Parts (for Ordnance Materiel)".

E–1b. No sand or shot blasting shall be used in connection with application of any finish unless specifically prescribed on the drawings.

E–2. *Components and assemblies.*—Components and assemblies shall conform to the dimensions within the tolerances, and shall be given the heat treatment and final or protective finish specified on the drawings.

E–3 *Barrel.*—

E–3a. All bearing surfaces shall be free from tool marks and other defects tending to prevent smooth operation. The barrel threads shall fit easily but not loosely in the barrel extension. Burrs or sharp edges shall be removed from the edge of the chamber.

E–3b. The enlargement of the chamber diameter due to proof-firing shall not exceed 0.001 inch, and there shall be no appreciable enlargement of the bore. The dimensions at these points after proof-firing shall be within the limits of the drawing.

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E–4. *Barrel extension.*—All working surfaces, particularly those on which the bolt, breech lock, and breech-lock cam operate, shall be smooth. The threading for the barrel shall fit easily but not loosely.

E–5. *Bolt.*—

E–5a. All operating surfaces shall be smooth and free from tool marks, burrs, and sharp edges. The T-slot shall be free from any roughness likely to collect brass particles.

E–5b. After proof-firing, the dimensions shall be as prescribed on the drawings and there shall be no set-back in the face of the bolt.

E–6. *Breech-lock.*—The breech-lock shall work smoothly but not loosely in its slot in the barrel extension.

E–7. *Casing with water jacket assembly.*—All working surfaces shall be smooth and free from tool marks and other defects tending to affect smoothness of operation. The feedway and guide cuts for the backplate shall be free from burrs and sharp edges. All rivets shall be tight and the riveting shall present a workmanlike appearance. Each water jacket assembly shall be water-tight at 30 pounds per square inch hydraulic pressure and shall withstand the test prescribed in paragraph F–10 without evidence of leakage. With the approval of the contracting officer, an air pressure test may be substituted for the hydraulic test. The water jacket shall be submerged in water and an internal air pressure of 50 to 60 pounds per square inch maintained while under test without evidence of leakage.

E–8. *Cocking lever.*—The lever shall retract the sear-engaging notch of the firing pin not less than 0.010 inch beyond the engaging point of sear to assure proper engagement.

E–9. *Driving spring rod assembly.*—The driving spring rod shall be straight, and the assembly shall move freely in the bolt.

E–10. *Extractor.*—All functioning surfaces shall be smooth and free from burrs and sharp edges. The extractor stud shall fit closely but move freely in its hole in the bolt.

E–11. *Firing pin assembly.*—The striker point shall be smooth and well rounded. There shall be no hand stoning of striker point. The sear-engaging notch shall have a fine finish and the forward edge shall be sharp and free from wire edge and burrs. The firing pin assembly shall move freely in the bolt.

E–12. *Sear.*—The sear shall work smoothly in its guide cuts in the bolt without tendency to bind or stick.

E–13. *Adjustments.*—

E–13a. The firing pin protrusion shall conform to the following requirement:

	Range
Firing pin protrusion (inch)	0.060–0.068

E–13b. *Trigger pull.*—The trigger pull shall not be less than 7 pounds nor more than 12 pounds, measured at the center of the bow of the trigger in a line at right angles to the axis of the bore in the plane passing through the axis of the bore parallel to the side plates. The measurement may be made with the back plate removed.

E–13c. *Timing.*—The timing of the gun shall be such that the firing pin cannot be released when the distance between the forward edge of the barrel extension and the trunnion block is 0.120 inch, and such

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that the firing pin can be released before this distance is reduced to 0.030 inch.

E–14. *Cycle rate.*—The rate of fire shall be not less than 450 shots per minute using U. S. Government Cal. .30 standard Service Ball Ammunition on the cal. .30 M1917A1 Mount. The normal rate may be expected to be from 450 to 600 shots per minute when mounted on the test stand.

E–15. *Critical springs.*—Springs designated as critical by the drawings or, in the absence of specific requirements on the drawing, by the inspector, shall be subjected to an endurance test by means of a gymnasticating device as described in paragraph F–8.

E–16. *Marking.*—

E–16a. Each machine gun, unless otherwise specified, shall be marked in the position and size prescribed on the drawings as:

Model, name and number.
Manufacturer's name and other marking.
Operating markings, such as "S" and "F."

Also, subject to acceptance,

Serial numbers (see specification No. 52–0–1).
Proof marks (see paragraph F–5).
Army Inspector's stamp (see paragraph F–16).
Ordnance Final acceptance stamp (see paragraph F–16).

E–16b. *Component identification.*—Components and assemblies, wherever practicable, shall bear an identifying mark of the manufacturer. A component or assembly shall bear a piece-mark only when called for on the drawing. If a special piece-mark drawing is furnished, only the items listed thereon shall bear piece-marks.

E–17. *Acceptance tests.*—Machine guns shall withstand the functioning, accuracy and endurance tests specified in section F. At least 10 percent of the machine guns shall be function fired on the cal. .30, M1917A1, Mount.

F. METHODS OF INSPECTION AND TESTS.

F–1. *Place of inspection.*—Unless otherwise specified, inspection and testing shall be done at the manufacturer's plant, with labor, facilities, and equipment furnished by the contractor except as otherwise provided for. The inspector may subject all or part of any lot or lots of components or assemblies to such inspection and gaging as he deems necessary to determine compliance with the requirements.

F–2. *Lot.*—A lot for inspection purposes, normally shall consist of not more than 500 machine guns.

F–3. *Visual inspection and operating tests.*—Each machine gun shall be visually inspected for completeness of manufacture, assembly, finish and workmanship. The chamber and bore shall be examined for rust, pits, powder fouling, burrs, and other defects. Each gun shall be operated by hand to ascertain that the final adjustments have been made to assure proper operation.

F–4. *Proof-firing.*—Each assembled machine gun and each spare barrel, barrel extension, and bolt shall be proof-fired in the presence of the inspector, or his representative, using one Government standard high-pressure proof cartridge. A special fixture shall be provided by

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the contractor for use in proof-firing barrels, barrel extensions, and bolts intended as spares.

NOTE.—Care is essential in proof-firing. Shake each proof cartridge to verify that powder is loose before using. If proof cartridges with caked powder are encountered, they shall not be used. In the event of a long hangfire, the action of the gun shall not be opened for about 20 seconds, and then shall be opened rapidly by means of a lanyard, using proper precautions for the safety of all personnel. Proof cartridges found defective shall be forwarded to Frankford Arsenal for examination.

F–5. *Proof-marking.*—The person doing the proof-firing shall place proof marks, in the location prescribed on the drawings, on each accepted casing assembly, barrel, barrel extension, and bolt at the time of proof-firing.

F–6. *Headspace.*—Each machine gun shall, after proof-firing and before final acceptance, be gaged to check for conformity to the headspace requirements of the drawings. Any machine gun not within the tolerances prescribed shall be rejected.

F–7. *Firing-pin protrusion.*—From each lot of machine guns, at least 10 percent shall be checked for conformity to the protrusion requirements. At the discretion of the inspector, this percentage may be increased.

F–8. *Gymnasticating test.*—

F–8a. At least 10 springs from each lot shall be tested by stressing for 20,000 cycles through the same range, and at approximately the same velocity and frequency, as in service.

F–8b. The contractor shall provide equipment, including suitable recording cyclometer, for making this test and shall record the conditions and results of each test.

F–9. *Interchangeability test.*—Ten machine guns, selected by the inspector, from each lot, shall be tested for interchangeability.

F–9a. Components and assemblies readily disassembled, as shown in the list to be furnished by the contracting officer, shall be disassembled from the weapons. Components of each kind shall be placed together and mixed. The machine guns shall be reassembled without fitting or altering any component in any way except that hand-fitting will be allowed on not more than two percent of parts provided that, as a result, no component or assembly is rendered unsuitable for assembly in other machine guns. The 10 assembled machine guns shall operate and function properly.

F–9b. At least two of the above reassembled machine guns, from which all components included as "Spare Parts" in the list to be furnished by the contracting officer have been removed, shall be assembled using components intended for use as spare parts. There shall be no hand-fitting, and the machine guns shall operate and function properly.

F–9c. At the option of the inspector, any or all machine guns assembled from interchanged components may be subjected to function and accuracy firing tests.

F–10. *Water jacket test.*—

F–10a. The contractor shall supply the necessary plugs, piping connections, and gages.

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F–10b. When tested with hydraulic pressure, each water jacket assembly shall be tested by maintaining the required pressure for a period of three minutes.

F–10c. When tested with air pressure, each water jacket assembly shall be tested for two minutes. Not less than the last 30 seconds of the submersion period shall be spent in observing for leaks.

F–11. *Acceptance test firing.*—

F–11a. *Ammunition.*—U. S. Government caliber .30 Standard Service Ball Ammunition shall be used in all acceptance test firing.

F–11b. *Adjustments.*—Adjustments for firing-pin protrusion, trigger pull, and timing, as prescribed in section E, shall be maintained at all times during acceptance test firing.

F–11c. *Lubrication.*—Oil for lubricating purposes shall be the standard oil issued for service use in the caliber .30 M1917A1 machine gun.

F–11d. *Function firing.*—Each gun shall be tested for functioning by firing at least 200 rounds as follows:

F–11d(1). One hundred rounds shall be fired, in bursts of various lengths, to tune up and test for proper functioning.

F–11d(2). One hundred rounds shall be fired to note indications of failure to operate correctly and smoothly, 50 rounds of which shall be fired as one continuous burst. The rate of fire may be checked in this test.

F–11e. *Accuracy.*—Approximately 10 percent of the guns in each lost shall be mounted in a fixed rest approved by the inspector, or on a standard machine gun tripod securely fastened to a suitable base, and fired for accuracy on a 100-foot range. All rounds of a ten-round burst, fired full machine gun action, shall group within a 3–inch circle.

F–11f. *Endurance test.*

F–11f(1). From each lot of 500 machine guns or less furnished on a contract, at least one gun which has successfully passed all other tests shall be selected and subjected to a 10,000 round endurance test. Water may be circulated through the water jacket during this test, or water may be added from time to time to the water jacket. Bursts shall not exceed 250 rounds each, and after the firing of each 250 rounds the gun shall be cooled by appropriate means.

F–11f(2). The gun may be cleaned and oiled after each 500 rounds or at the close of the day's firing, but no parts shall be altered or replaced except that broken parts, or parts worn to the extent that they are unserviceable, shall be replaced. A record shall be kept of each replacement.

F–11f(3). Upon completion of the endurance test, the gun shall be inspected and all defective or worn components replaced. Before acceptance, the gun shall be thoroughly conditioned at the expense of the contractor.

F–11f(4). To be acceptable, guns shall withstand the endurance-firing test without substitution of any components, without breakage or excessive wear of any component, and without malfunctions, not directly traceable to defective amunition, in excess of the limits shown in table I.

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F–12. *Malfunctions, broken parts and nonacceptable conditions.*— In any test except the endurance test, malfunctions, broken parts and nonacceptable conditions, including those listed in table I, shall cause rejection of the gun being tested.

TABLE I.—*Malfunctions, broken parts, and nonacceptable conditions*[1]

Malfunctions and nonacceptable conditions	Number permitted in the endurance test	
	First 5,000 rounds	Second 5,000 rounds
Enlarged firing pin hole	0	0
Extruded primers (enlarged firing pin hole)	0	2
Failure of cover latch	0	1
Failure to close (due to any gun component)	0	1
Failure to eject	1	2
Failure to extract (cartridge case from chamber)	0	1
Failure to extract (cartridge from feed belt)	0	1
Failure to feed (bullet striking face of barrel)	0	2
Failure to feed (insufficient recoil)	0	1
Failure of trigger to release	0	1
Faulty trigger pull	0	[2] NP
Hangfire (noticeable)	1	0
Misfire due to gun	0	2
Pierced primer	0	1
Partial ignition	0	0
Ruptured cartridge case	0	1
Uncontrolled fire	0	0
Other malfunctions, total	3	5
Total malfunctions	5	21
BROKEN OR UNSERVICEABLE PARTS		
Accelerator, broken	0	0
Barrels (eroded)	0	[2] NP
Barrel, extension, eroded	0	0
Bolt, broken	0	0
Cocking lever, broken	0	0
Cams, broken	0	0
Leaky water jacket	0	0
Leaks in end cap or trunnion due to wear	0	0
Cover pin, broken or burred	0	1
Ejector, broken	0	0
Extractor assembly, broken, burred, or deformed	0	0
Extractor cam, broken or burred	1	1
Firing pin, broken	0	1
Firing pin spring, broken or set	0	2
Rivets, loose	0	0
Belt feed pawl, broken or burred	0	0
Belt feed slide, broken or burred	0	1
Sear, broken	0	1
Sear contacts, defective	0	0
Side or bottom plates, spread or broken	0	[2] NP
Striker point, defective	0	1
Set back in face of bolt	0	8
All other broken parts		
Total unserviceable parts	1	16

[1] See par. F–13.
[2] Indicates no penalty if encountered in the endurance test.

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F–13. *Defective ammunition.*—Malfunctions in any test traceable to defective ammunition shall not be counted against the machine gun being tested.

F–14. *Retests.*—

F–14a. Machine guns rejected individually because of defective components or assemblies in any inspection or test except the endurance test may be conditioned and resubmitted for the inspection or test in which failure occurred, and for such other tests as the inspector may consider necessary, provided the number of machine guns submitted for such retests does not exceed two percent of the lot. Retests of a larger number may be made at the discretion of the inspector.

F–14b. If the machine gun representing any lot fails to meet the specified requirements in the endurance test a retest shall be made unless, in the opinion of the inspector, the failure or failures indicate serious defects in the lot of machine guns, in which case retest shall be made only if authorized by the contracting officer and after the entire lot has been conditioned where necessary. In case a retest is made, the inspector shall select another gun for the purpose from the lot under consideration. Failure of the gun in the retest shall be cause for rejection of the lot.

F–15. *Final inspection.*—Machine guns which pass all the required tests shall be thoroughly cleaned and then given such visual inspection as the inspector deems necessary.

F–16. *Marking.*—Each accepted machine gun shall be stamped, adjacent to the manufacturer's marking on the right hand side-plate, with the inspector's stamp and with the Ordnance final inspection stamp (circle, crossed guns, and flaming shell, approximately one-fourth inch in diameter).

G. PACKING AND MARKING.

G–1. *Preparation.*—After acceptance of machine guns and spare parts, the contractor shall prepare them for shipment. Machine guns shall be slushed with a suitable rust-preventive compound conforming with U. S. Army Specification No. 2–84 before packing. Spare parts shall be coated with a rust-preventive compound conforming with Specification No. 2–84 and then wrapped in grease-proof paper conforming with Ordnance Department Tentative Specification AXS–840. Grade A grease-proof paper shall be used for parts not having a protective finish but having a coating of rust-preventive. Grade B grease-proof paper shall be used for assemblies and parts having a protective finish and a coating of rust-preventive .

G–2. *Packing.*—

G–2a. *Machine guns.*—Machine guns shall be packed in accordance with applicable drawings, specifications, and Ordnance Packaging Instructions P. S. No. 21.

G–2b. *Spare parts.*—Spare parts shall be packed in quantities approximately proportional to the quantities supplied per lot of machine guns. Unles otherwise specified, packing shall be in accordance with good commercial practice and such as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Packing construction shall be adequate for stacking in

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storage and shall be such that any container or box shall withstand dropping from a height of five feet without rendering the container unserviceable or damaging the contents.

G–3. *Marking.*—Marking shall be in accordance with U. S. Army Specification No. 100–2.

H. NOTES.

H–1. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

H–2. This specification, if applicable, may be used in the procurement of machine guns not listed in paragraph B–1.

NOTICE.—This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced, either wholly or in part, except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

NOTE.—Copies of this specification may be obtained from the Office of the Chief of Ordnance, War Department, Washington, D. C.

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U. S. ARMY
SPECIFICATION

No. 52-5-1B
4 June 1943
Amendment-2
28 July 1944
Superseding
Amendment-1

GUN, MACHINE, BROWNING, CALIBER .30, M1917A1

The following changes shall be made in Specification No. 52-5-1A, dated 4 June 1943:

Amendment-1 dated 7 April 1944 is hereby cancelled.

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